SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1108

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models**.** *(BL4)*

Assessment Tool Weightage: 40%

Code of Conduct:

I E.Ragul / 192571032 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E.Ragul

Case study

QUESTIONS: Any 4 **Total Marks:40**

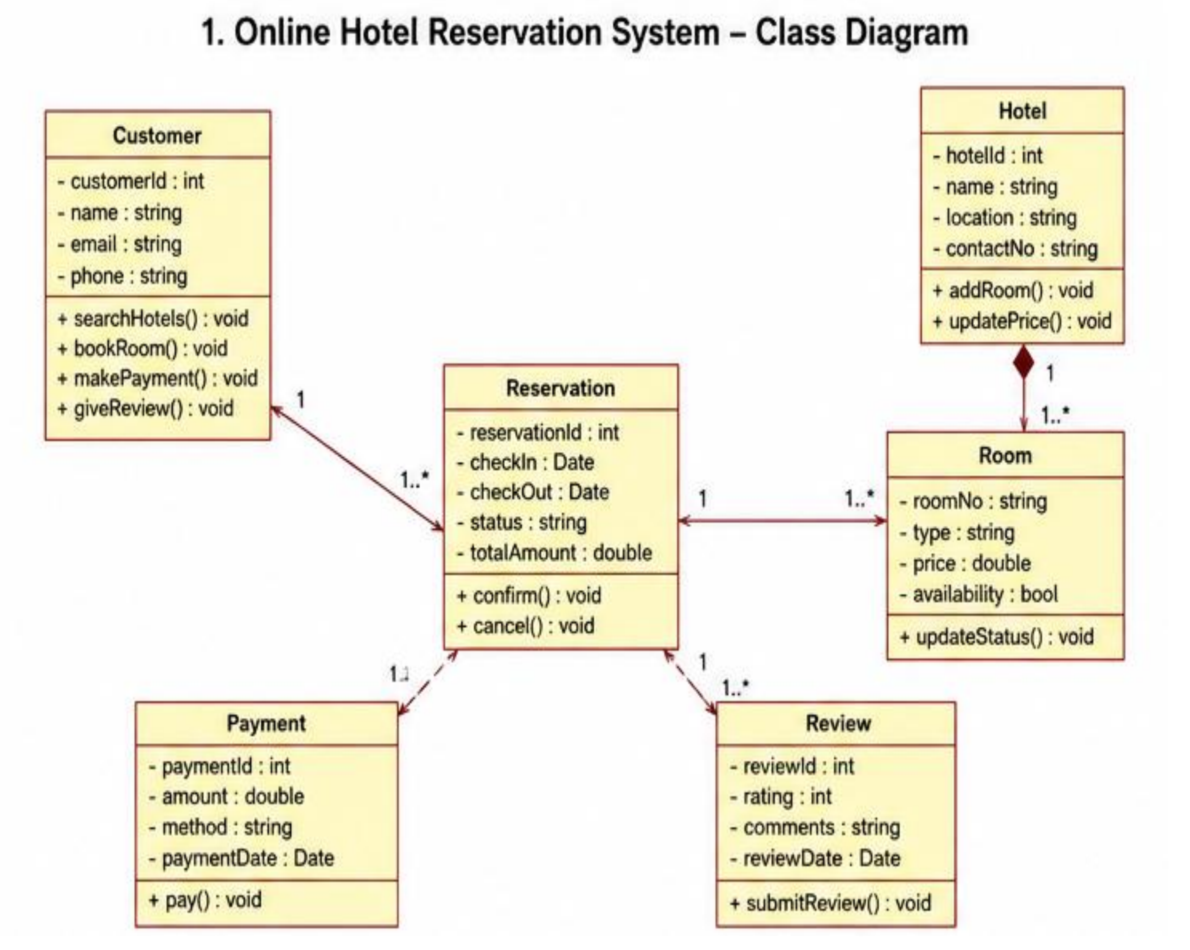
1. Online Hotel Reservation System

A travel company plans to develop an online hotel reservation platform where customers can search hotels, book rooms, make payments, and provide reviews. Hotel managers can update room availability and pricing, while administrators manage users and transactions.

Question:

Analyze the system and explain how object-oriented principles (encapsulation, inheritance, polymorphism) can be applied. Also, suggest a suitable SDLC model and justify your choice for this system.

ANSWER:



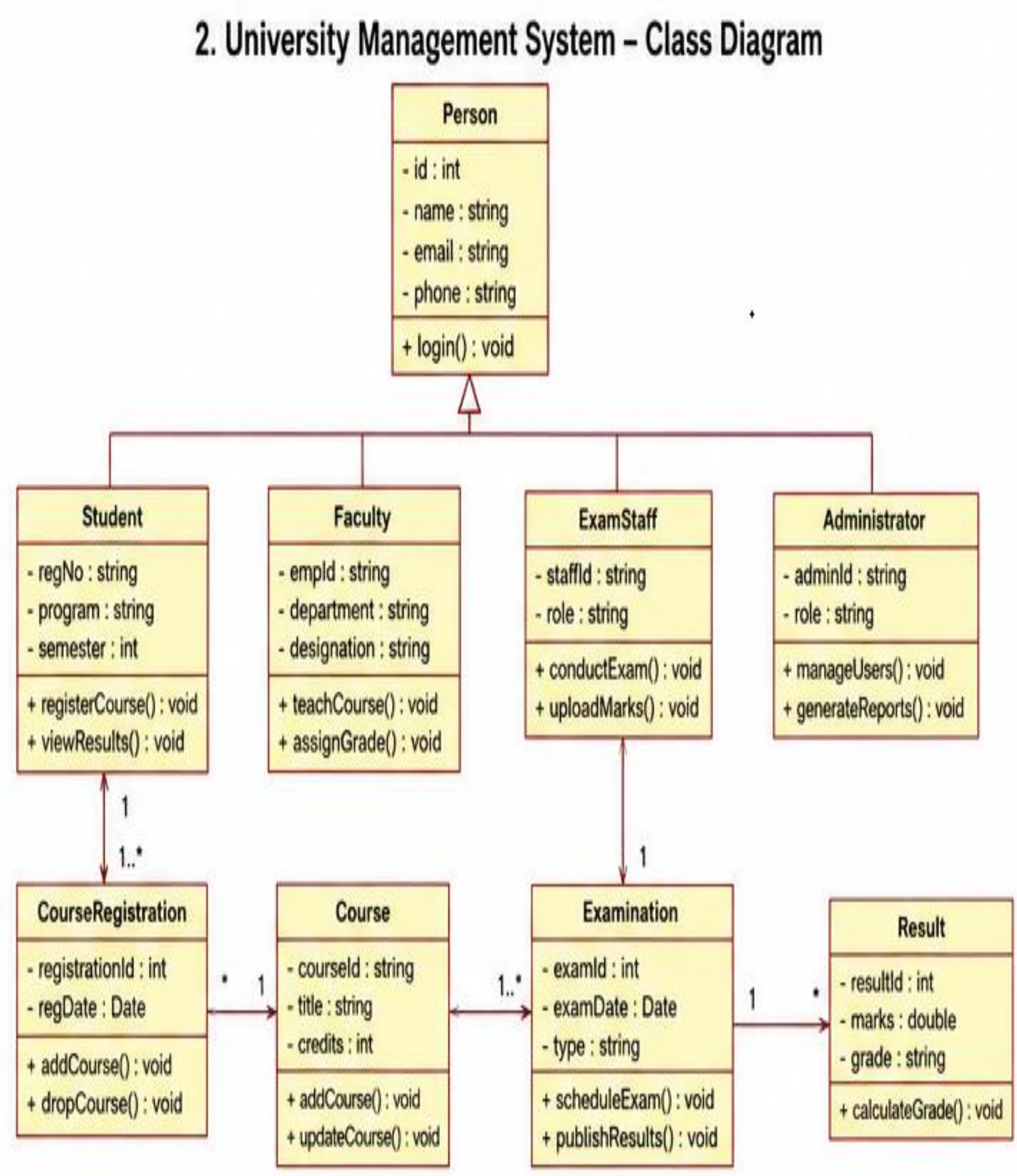
2. University Management System

A university intends to automate its operations including student admissions, course registration, examination management, and result processing. Different users such as students, faculty members, examination staff, and administrators interact with the system.

Question:

Design a structured approach using object-oriented concepts to model this system. Identify key classes and relationships, and recommend an SDLC model for efficient development with reasoning.

ANSWER:



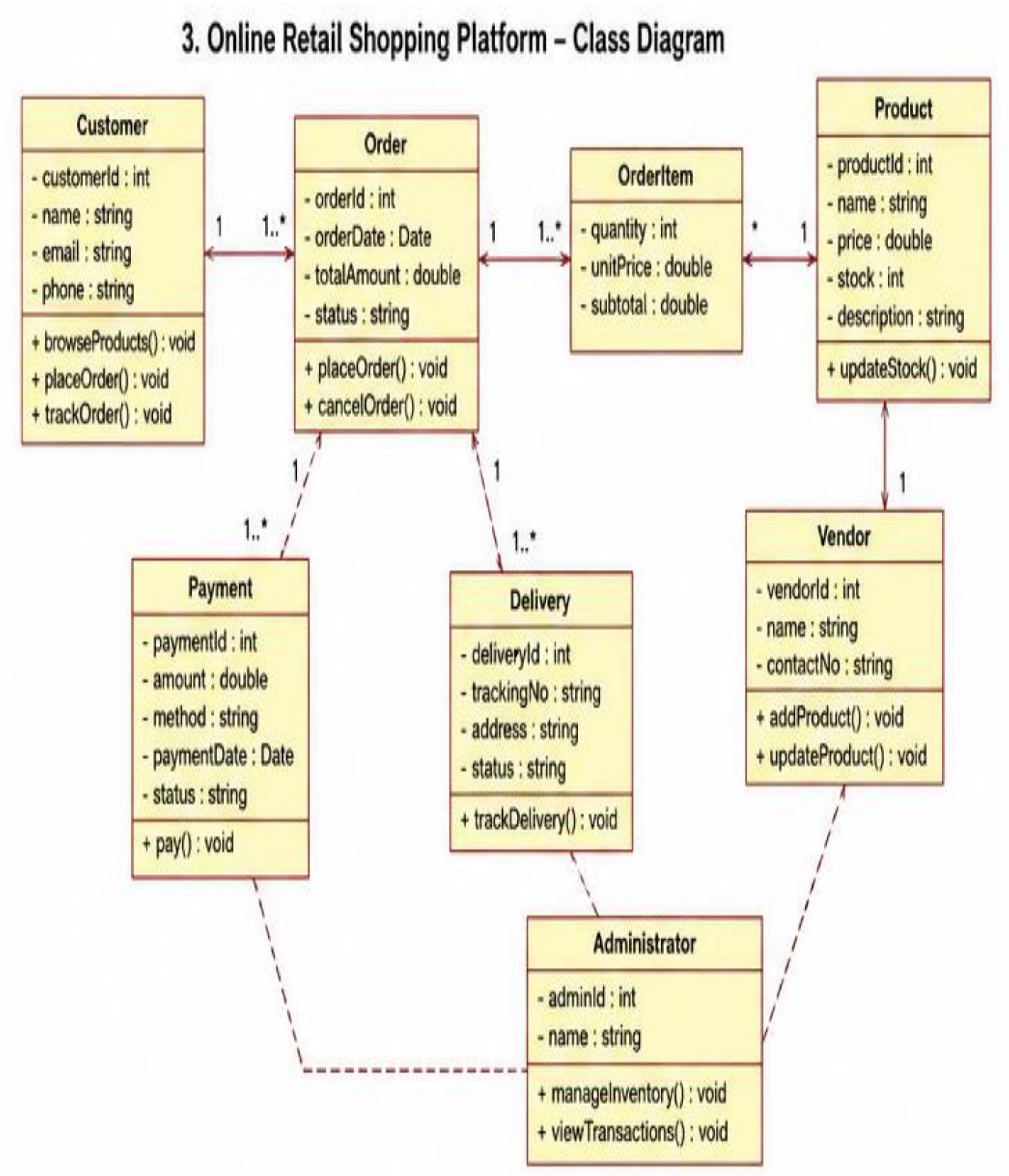
3. Online Retail Shopping Platform

A retail company wants to develop an e-commerce platform where customers can browse products, place orders, track deliveries, and make online payments. Vendors manage product catalogs, while administrators monitor transactions and inventory.

Question:

Discuss how modular software design can be achieved using object-oriented techniques. Evaluate different SDLC models and select the most appropriate one for this scenario with justification.

ANSWER:



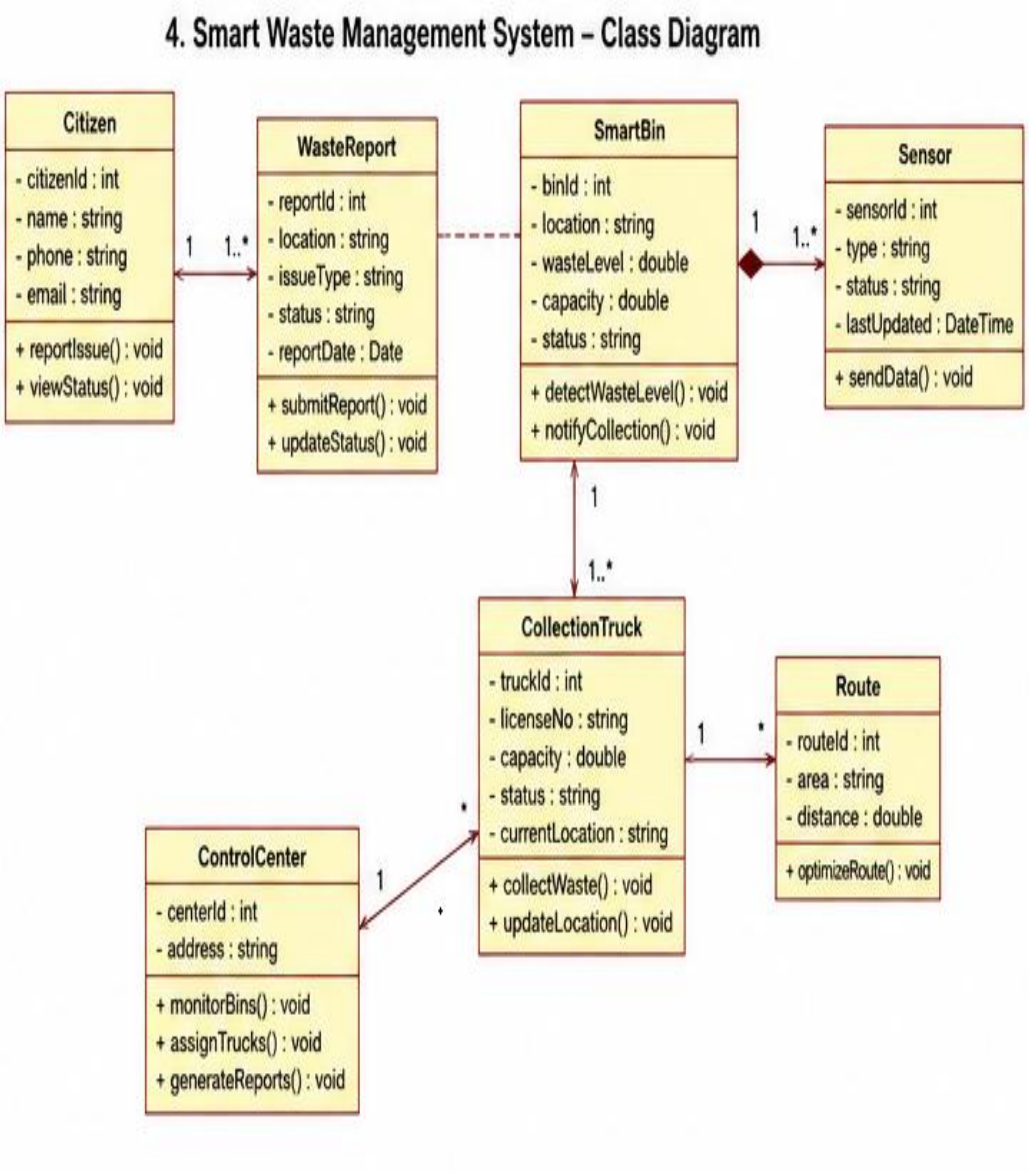
4. Smart Waste Management System

A municipality introduces a smart waste management system where IoT-enabled bins monitor waste levels and automatically notify collection vehicles when bins are full. Citizens can also report waste-related issues through a mobile application.

Question:

Apply object-oriented system development principles to design this system. Identify the life cycle model best suited for handling real-time data, sensor integration, and evolving requirements, and explain why.

ANSWER:



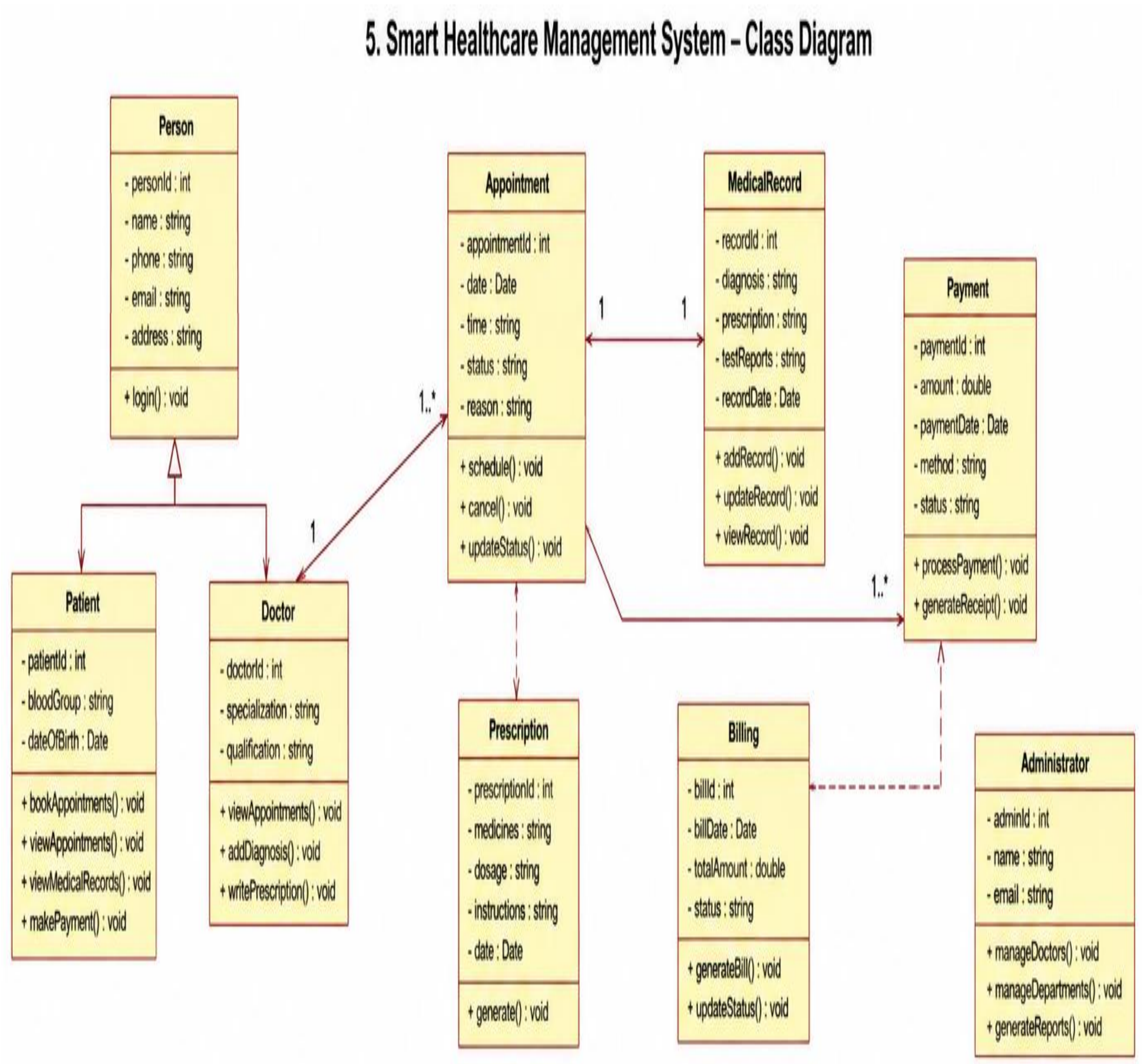
5. Smart Healthcare Management System

A multi-specialty hospital plans to develop a Smart Healthcare Management System where patients can book appointments, consult doctors, access medical records, and make online payments. Doctors update patient diagnoses and prescriptions, while hospital administrators manage departments, staff, and billing information.

Question:

Analyze the system and explain how **object-oriented principles (encapsulation, inheritance, and polymorphism)** can be applied. Also, recommend a suitable **SDLC model** for developing this healthcare system and justify your choice.

ANSWER:



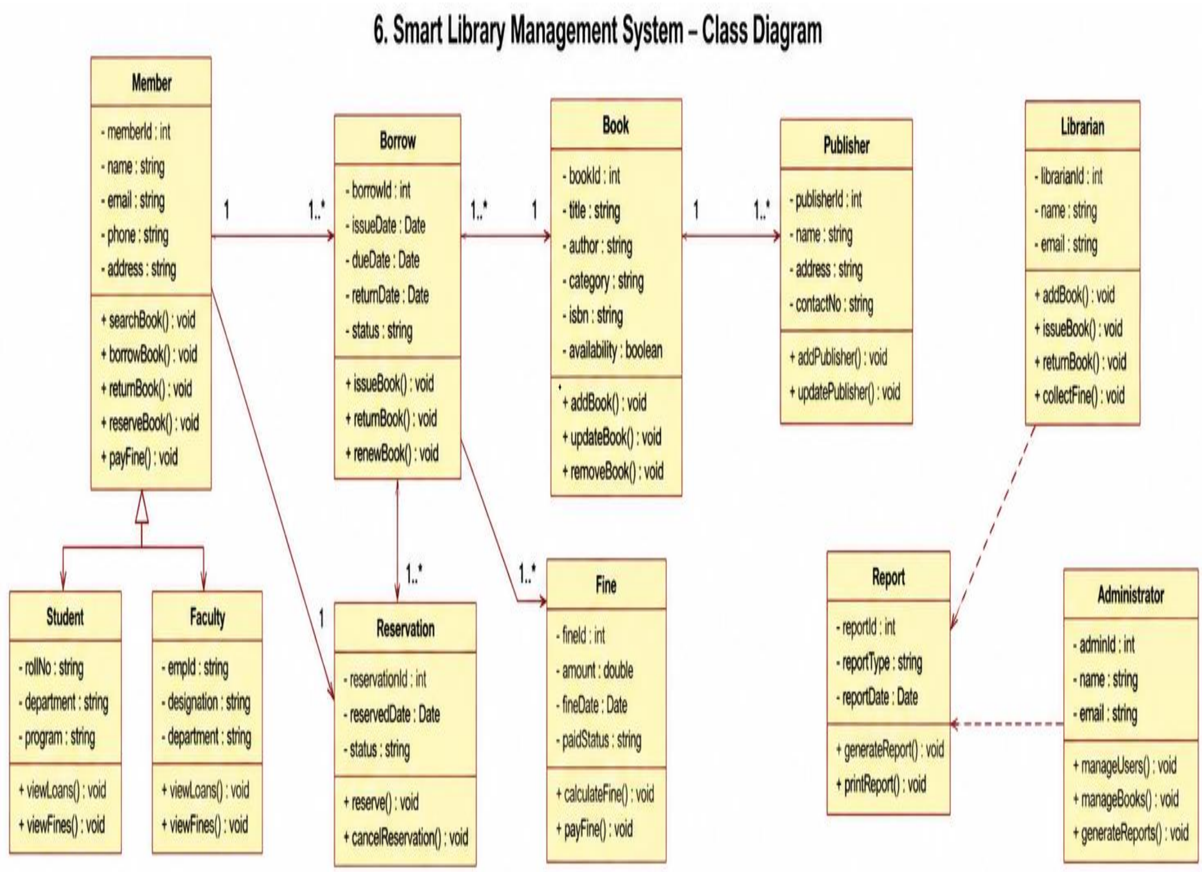
6. Smart Library Management System

A university library intends to automate its operations, allowing students and faculty members to search books, borrow and return books, reserve titles online, and pay overdue fines. Librarians manage book inventories, memberships, and circulation records, while administrators generate reports and maintain the system.

Question:

Design a structured solution using **object-oriented concepts** for this system. Identify the major classes, attributes, methods, and relationships, and recommend an appropriate **SDLC model** for efficient development with suitable justification.

ANSWER:



RUBRICS FOR CALCULATION:

Case study

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand